Swati Gavas

Personal Data

Place and Date of Birth:	Kolhapur, Maharashtra, India | July 1994
Nationality and Gender:	Indian | Female
Webpage:	swati-gavas.github.io/profile/
Phone and email:	+91 7021983364 | swatigavas47@gmail.com

Post-Doctoral Positions

Tenure:	APR 2025 - Present
Advisor/Host:	Dr. Nishikanta Khandai
Institute:	National Institute of Science Education and Research (NISER), Bhubaneswar

PhD

Tenure:	AUG 2018 - DEC 2024
Topic:	Aspects of gravitational clustering and structure formation in the Universe
Advisor:	Prof. Jasjeet Singh Bagla
Institute :	Indian Institute of Science Education and Research (IISER), Mohali

Education

July 2017 — Master of Science in Physics,
Department of Physics, Savitribai Phule Pune University, Pune
Major: Astronomy and Astrophysics | CGPA: 8.27/10

July 2015 — Bachelor of Science in Physics
DBJ College, Chiplun, **Mumbai University** | CGPA: 7/7
Secured third rank at the University level

Research Interest

- Gravitational clustering and large-scale structure formation through cosmological simulations and analytical modeling, linking theory with survey observations.
- Dark matter halos, galaxy formation, and baryonic processes

Publications NASA-ADS

2022 : Swati Gavas, Jasjeet Bagla, Nishikanta Khandai, Girish Kulkarni
Halo mass function in scale invariant models,
MNRAS, Volume 521, Issue 4, June 2023, Pages 5960–5971

2024 : Swati Gavas, Jasjeet Bagla, Nishikanta Khandai
Dispersion in the Hubble-Lemaître constant measurements from gravitational clustering,
Phys. Rev. D 111, 043516 – 10 February, 2025

Jasjeet Bagla, Swati Gavas
On the origin of transient features in cosmological N-Body Simulations,
JOAA, Volume 46, article number 33, 16 June 2025

2025 : [Submitted]
Dipayan Mukherjee, Harkirat Singh Sahota, Swati Gavas
A dynamical systems perspective on the thermodynamics of late-time cosmology
gr-qc, astro-ph.CO

2026 : [In preparation]
Ayan Nanda, Nishikanta Khandai, Jasjeet Singh Bagla, Swati Gavas
Self-Similarity of Halo Shapes in Cosmological Simulations.

[In preparation]
Swati Gavas, Ayan Nanda, Nishikanta Khandai
Finite box size effect on halo scales

[In preparation]
Ayan Nanda, Nishikanta Khandai, Swati Gavas
Multipole Characterization of Triaxial Dark Matter Halos:
A Spherical Harmonic Approach

Conferences, Schools and Workshops

29-2 Oct 2025: (presented talk)
Challenges and Innovations in Computational Astrophysics VI
Held at Indian Institute of Science Education and Research (IISER), Mohali
Talk title: Numerical Artifacts of Restricted Power Spectrum in cosmological N-Body Simulations

11 Nov 2024: (presented talk)
Computational and Observational Cosmology Group at the Observatório Nacional - ON(RJ/Brazil)
Talk title: Dispersion in the Hubble-Lemaître constant measurements from gravitational clustering

14-25 Oct 2024: **3rd IAGRG School on Gravitation and Cosmology**
Held at International Centre for Theoretical Sciences (ICTS), Bengaluru, India

1-2 July 2024: (presented talk)
Séminaire Univers Institut d'Astrophysique de Paris (IAP)
Visit to Prof. Stephane Colombi at Institut d'Astrophysique de Paris (IAP), France
Talk title: Halo mass function in scale-invariant models

17-28 June 2024: (presented talk)
Summer School on Cosmology
Held at International Centre for Theoretical Physics (ICTP), Trieste, Italy
Talk title: Dispersion in the Hubble-Lemaître constant measurements from gravitational clustering

1-4 Feb 2024: (presented talk)
42nd Meeting of Astronomical Society of India
Held at IISc, ISRO and JNP Bengaluru, India
Talk title: Dispersion in the Hubble-Lemaître constant measurements from gravitational clustering

6-9 Dec 2023: (presented talk)
10th International Conference on Gravitation and Cosmology
Held at IIT Guwahati, India
Talk title: A Local Perspective on Hubble Tension from Cosmological N-body Simulations

7-9 Nov 2023: (presented talk)
Challenges and Innovations in Computational Astrophysics V
Virtual Meeting
Talk title: A Local Perspective on Hubble Tension from Cosmological N-body Simulations

24-28 Oct 2022: (presented talk)
The 10th KIAS Workshop on Cosmology and Structure Formation
Held at KIAS, South Korea
Talk title : Halo mass function in scale invariant models

Fellowships, Scholarships & Awards

ANRF-ITS 2024: International travel grant to attend school in Italy
INSA-INYAS SciArt 2021: International SciArt Image Competition 2021
Third prize in simulation category for first entry
Certificate of appreciation for second entry
UGC-SET 2018: For lectureship in physics
INSPIRE Fellowship 2017: Fellowship for Doctorate program in physics
GATE physics 2018: All India Rank- 484
JAM physics 2015: All India Rank- 460
INSPIRE SHE: INSPIRE Scholarship for Higher Education
Tenure: 2012-2017

Other Research Experience

Jan-May 2017 **Masters project at IUCAA, Pune, India**
N-body simulations to study large scale structure of the universe and study halo properties.
Supervisor : Dr. Aseem Paranjape, IUCAA, Pune, India.

jun-july 2016 **Summer School Project at IPR, Gandhinagar, India.**
Time dependent electron-ion collision frequency in a strong laser field using non-Maxwellian electron velocity distribution function
Supervisor : Dr. Mrityunjay Kundu, IPR, Gandhinagar, India.

Teaching & Mentoring

Jan-present 2025: Co-mentoring a PhD student as a part of collaboration
Aug-Dec 2022: PHY111 Physics lab Mechanics (IISER Mohali)
Jan-May 2021: IDC201 Introduction to astronomy (IISER Mohali)
Jan-May 2020: PHY212 Modern Physics Lab (IISER Mohali)
Aug-Dec 2019: PHY411 Nuclear Physics Lab (IISER Mohali)

Technical Skills

High Performance Computing :	Kalinga(NISER), astro (NISER), HPC-IISERM
Programming Languages :	Python, Julia, Fortran90, C/ C++.
Operating Systems :	Linux based systems
Typesetting software :	LaTeX

Languages

Mother tongue:	Marathi
Fluent:	English, Hindi

OTHERS

- Dec 2009: **National Means Cum-Merit Scholarship Scheme (NMMSS)**
  Tenure: 2009-2012